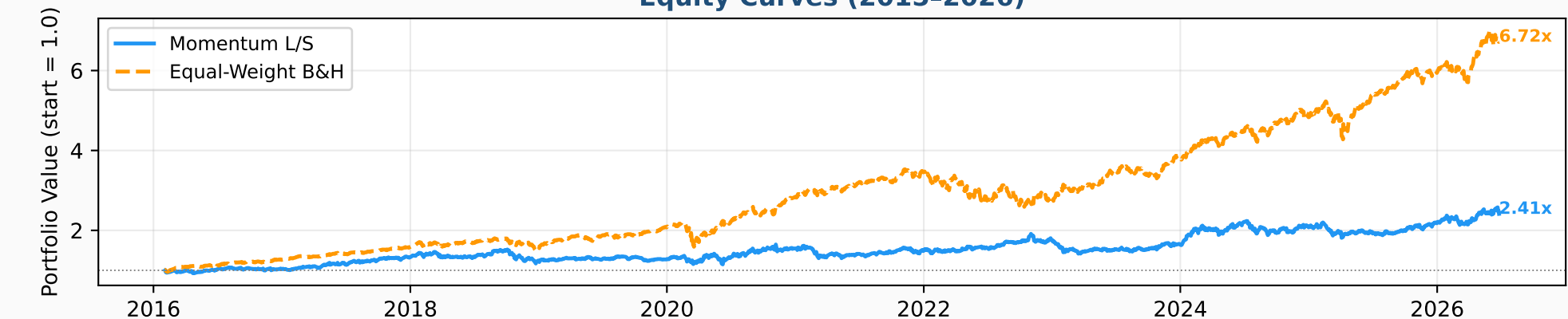


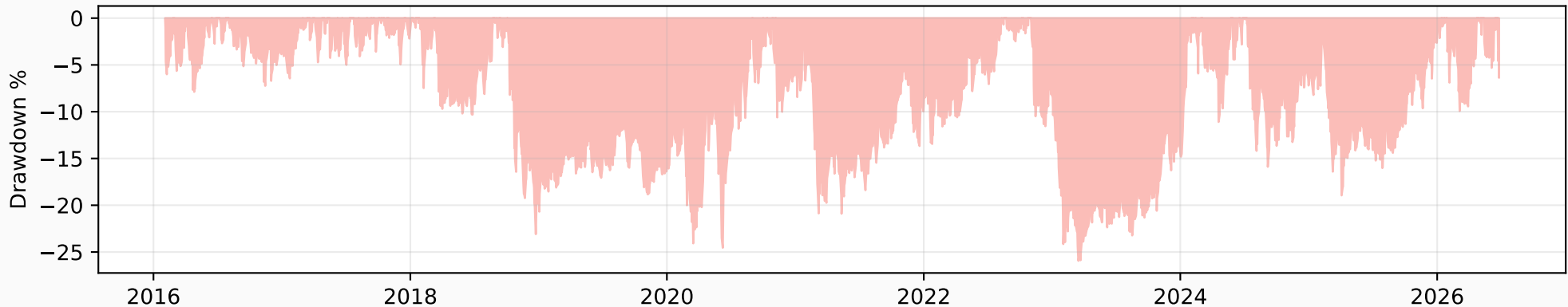
QuantAlpha — Systematic Trading Tearsheet

Diego Mella Valerio | MSc Financial Engineering, UAI | June 2026

Equity Curves (2015-2026)



Strategy Drawdown



Performance Summary

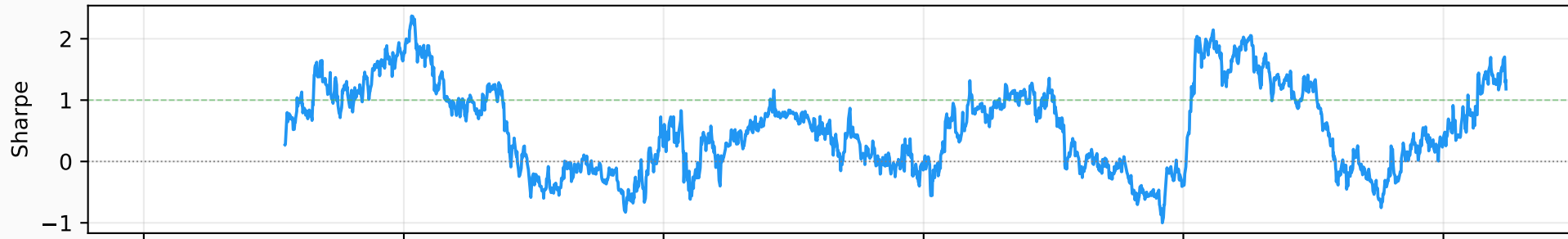
	Momentum L/S	Equal-Weight
CAGR	8.87%	20.16%
Sharpe	0.54	1.18
Max DD	-25.94%	-28.20%
Ann. Vol	19.08%	16.70%
Calmar	0.34	0.71

Tail Risk (Daily)

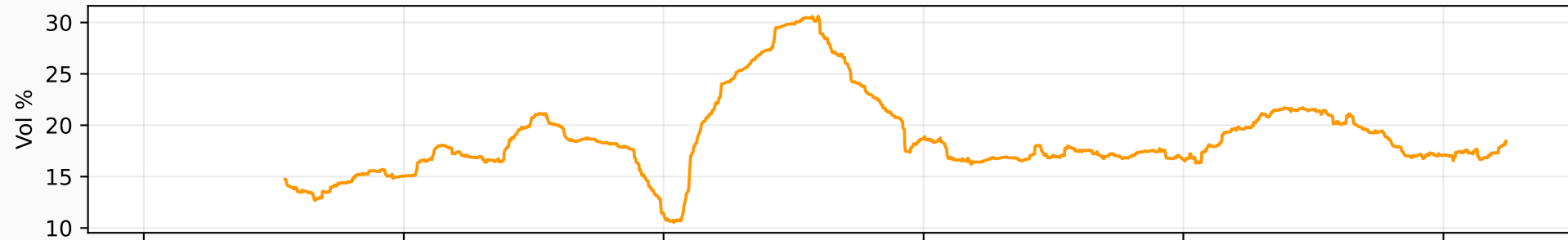
	Value
VaR 95%	-2.00%
VaR 99%	-3.43%
CVaR 95%	-2.89%
CVaR 99%	-4.31%
Skewness	-0.422
Kurtosis	2.780

QuantAlpha — Rolling Risk Metrics (1Y Window)

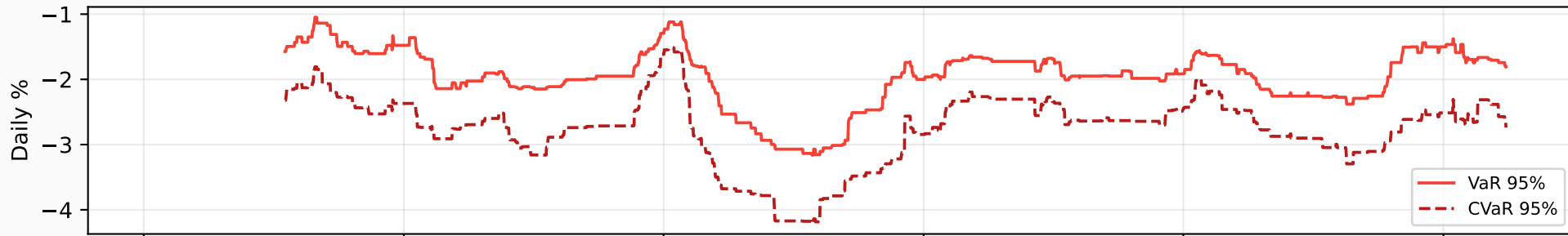
Rolling 1Y Sharpe Ratio



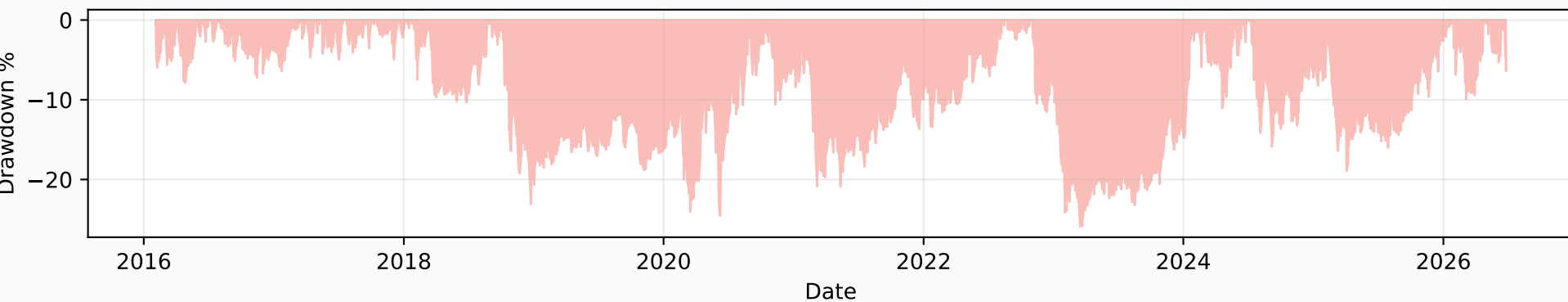
Rolling 1Y Annualized Volatility



Rolling 1Y VaR & CVaR (95%)

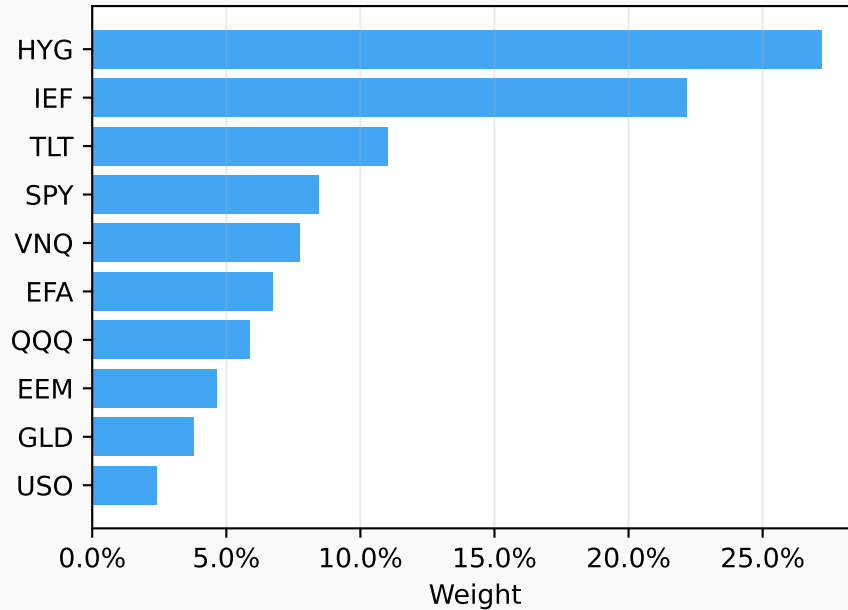


Drawdown

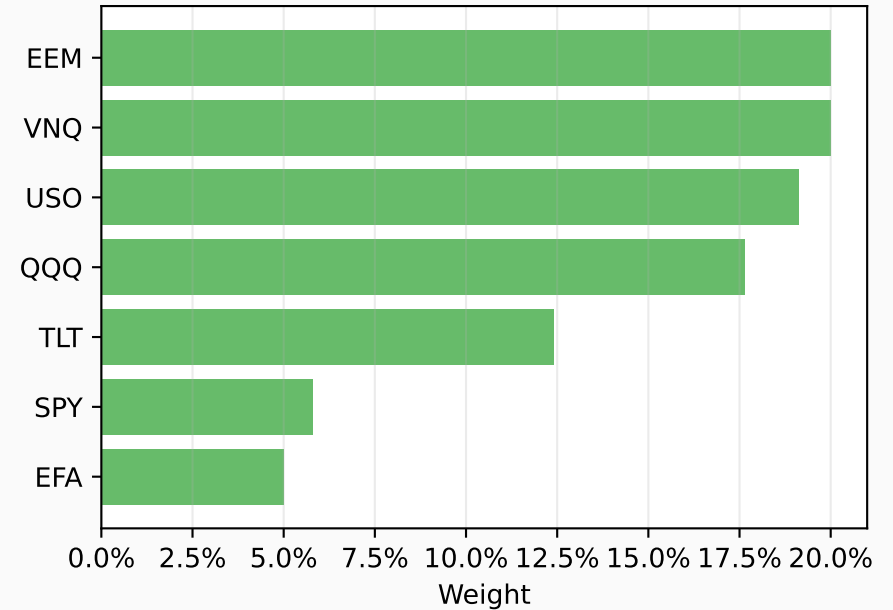


QuantAlpha — Portfolio Optimization

Risk Parity Weights
(top 10, inverse vol)



Max Sharpe Weights
(mean-variance, long-only)



Weight Comparison

	Risk Parity	Max Sharpe
SPY	8.4%	5.8%
QQQ	5.9%	17.6%
GLD	3.8%	0.0%
TLT	11.0%	12.4%
EEM	4.6%	20.0%
EFA	6.7%	5.0%
HYG	27.2%	0.0%
VNQ	7.7%	20.0%
USO	2.4%	19.1%
IEF	22.2%	0.0%

RISK PARITY
Weight inversely proportional to vol. Lower-vol assets (bonds, gold) receive more weight. Equal risk contribution. Used by: Bridgewater All Weather.

MAX SHARPE (MEAN-VARIANCE)
Maximizes return per unit of risk. Sensitive to return estimates – tends to concentrate in recent winners. Known limitation: estimation error.

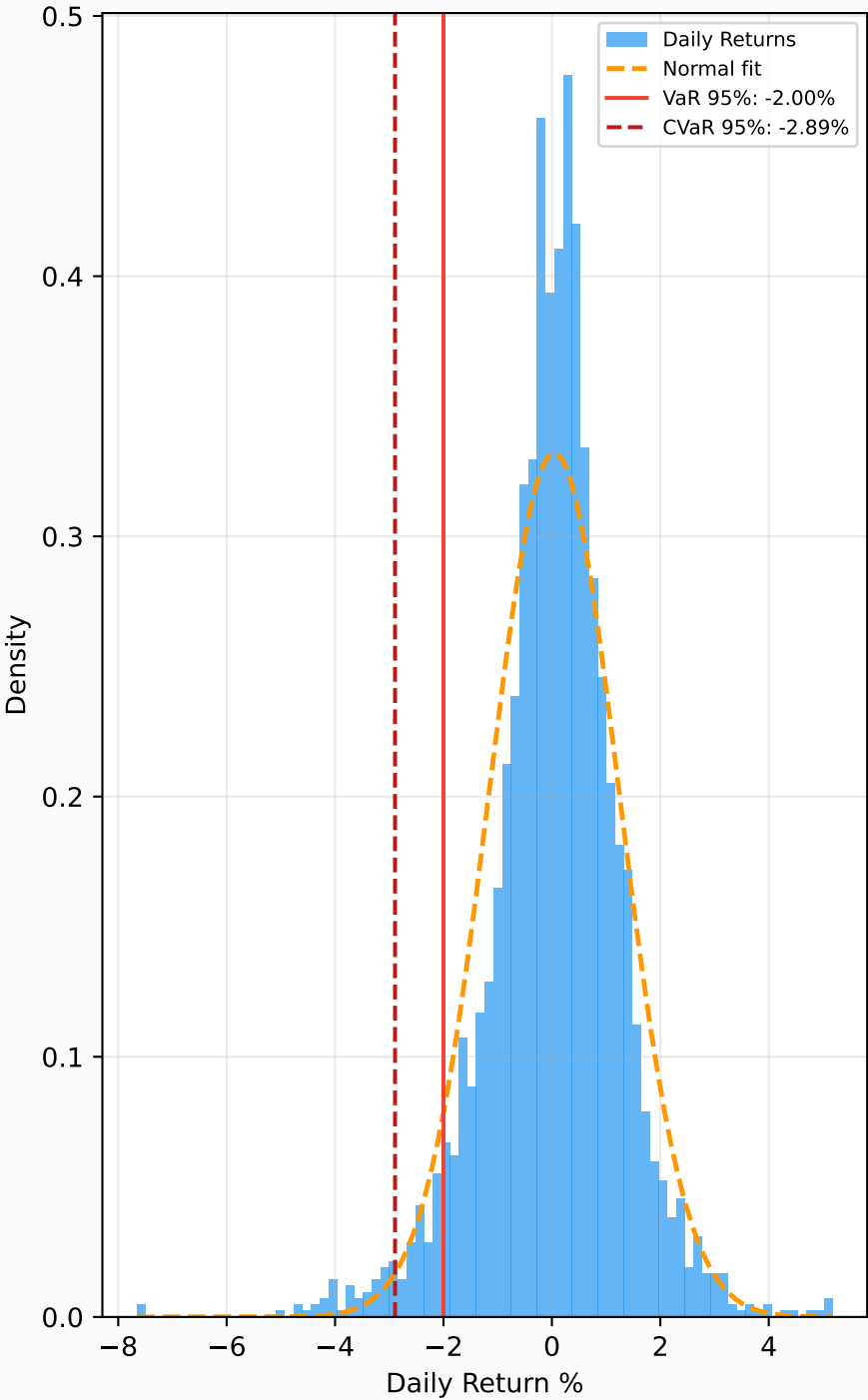
KNOWN LIMITATION
Both optimizers use in-sample data. Out-of-sample performance degrades due to estimation error. Production use requires rolling reestimation.

QuantAlpha — Full Risk Report

Complete Risk Metrics

	Momentum L/S
— RETURN METRICS —	
Total Return	141.38%
CAGR	8.87%
Ann. Volatility	19.08%
Sharpe Ratio	0.54
— DRAWDOWN —	
Max Drawdown	-25.94%
Calmar Ratio	0.34
— TAIL RISK (daily) —	
VaR 95% Historical	-2.00%
VaR 99% Historical	-3.43%
VaR 95% Parametric	-1.94%
VaR 95% Cornish-Fisher	-2.01%
CVaR 95% (Exp. Shortfall)	-2.89%
CVaR 99% (Exp. Shortfall)	-4.31%
— DISTRIBUTION —	
Skewness	-0.422
Excess Kurtosis	2.780
Best Day	5.17%
Worst Day	-7.65%
% Positive Days	53.6%

Return Distribution vs Normal



QuantAlpha — Analysis & Interpretation Guide

What every chart and metric means, and what this strategy's numbers tell us

PAGE 1 — EQUITY CURVES & PERFORMANCE TABLES

Equity Curve

Shows how \$1 invested in Jan 2015 grew over time. Momentum L/S reached 2.41x vs 6.72x for the equal-weight benchmark.

The gap is explained by the benchmark's heavy exposure to NVDA, META and AAPL which compounded exceptionally over 2015-2026.

The momentum strategy is market-neutral (long + short), so it does not fully participate in bull markets but also limits downside.

Strategy Drawdown

Each trough shows how far the portfolio fell from its previous peak. Worst drawdown: -25.94% vs benchmark -28.20%.

The momentum strategy's shallower drawdown confirms its risk management value through short positions in underperformers.

The deepest trough (2019-2020) reflects the COVID momentum crash: markets reversed faster than the monthly signal could adapt.

Performance Table

CAGR 8.87% vs 20.16% benchmark. Against SPY alone (~12% CAGR), the gap is much smaller.

Sharpe 0.54: for every unit of risk taken, the strategy earned 0.54 units of excess return (annualized).

Calmar 0.34: CAGR divided by max drawdown. Higher = better return per unit of worst-case loss.

Tail Risk Table

VaR 95% -2.00%: on 95% of days the strategy loses less than 2.00%. This is the daily risk budget.

CVaR 95% -2.89%: on the worst 5% of days, average loss is 2.89%. Always worse than VaR - captures tail severity.

PAGE 2 — ROLLING RISK METRICS (1Y WINDOW)

Rolling Sharpe

Shows how risk-adjusted performance evolved year by year. Green dashed line = Sharpe 1.0 (institutional target).

Peaks above 1.5 in 2017 and 2024 = strong momentum regimes. Troughs below 0 in 2020 and 2022 = momentum crashes.

The 2020 crash: COVID caused a sharp reversal that punished momentum strategies globally. The 2022 crash: rate hikes.

Rolling Volatility

Spike to ~30% in 2020 shows how risk surged during COVID. Normal range is 15-22%. Currently ~18%.

Rolling VaR & CVaR

Both widened sharply in 2020 (CVaR reached ~-4%). Shows the risk model correctly detected the regime change in real time.

The gap between VaR and CVaR measures tail severity - wider gap = fatter tails beyond the VaR threshold.

PAGE 3 — PORTFOLIO OPTIMIZATION

Risk Parity Weights

HYG and IEF receive the highest weights because they have low volatility. USO receives the least (highest vol commodity).

This is Bridgewater's All Weather logic: allocate risk equally, not capital equally.

Max Sharpe Weights

Concentrated in EEM, VNQ, USO, QQQ - assets with high recent Sharpe ratios. Uses in-sample data so subject to overfitting.

Note GLD and HYG get 0% - the optimizer found their historical Sharpe insufficient given correlation to other assets.

PAGE 4 — RETURN DISTRIBUTION

Distribution vs Normal Fit

The blue histogram shows actual returns. The orange dashed curve is a normal distribution fitted to the same mean/vol.

Key observation: the actual distribution has a taller, narrower peak and fatter tails than the normal. This is leptokurtosis.

Excess kurtosis 2.78 (normal = 0) confirms fat tails. Skewness -0.422 confirms slight negative skew (losses > gains).

Practical implication: parametric VaR (which assumes normality) underestimates tail risk. This is why we also compute

Cornish-Fisher VaR which adjusts for skewness and kurtosis, and historical VaR which uses actual return data.